

Project Proposal — Programming AI Accelerators (CS3.406)

1. Project Overview

Project Title: Improving Inference Speed and Energy Efficiency of a Vision-Language Model

Team Members:

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Project Direction: Applications & Edge AI

Brief Problem Statement & Motivation:

Vision-language models are being used more and more in real-world applications such as robots, smart cameras, and visual assistants. In many of these use cases, the model needs to respond quickly and without using too much power, especially when running on smaller or edge devices. However, running these models efficiently is still a hard problem, and it is often not clear where exactly the time and energy are being spent during inference. In this project, we want to study how a vision-language model performs during inference, find out where the actual slowdowns are, and try to make it faster and more energy efficient using GPU-level optimizations.

2. Proposed Work

What we will build, implement, or optimize:

We plan to work with Qwen2-VL, an open-source vision-language model. The code for this model is publicly available here: <https://github.com/QwenLM/Qwen2-VL>.

Our plan is to:

- Run and study the model's inference process end to end.
- Profile the inference to understand where time and energy are actually being spent.
- Identify the main bottlenecks that are slowing things down or using unnecessary energy.
- Try to optimize these bottlenecks, mainly using GPU/CUDA-level techniques.

Key technical approach / techniques (high level):

We plan to use standard profiling tools such as the PyTorch Profiler and NVIDIA's Nsight Systems and Nsight Compute to understand the model's behaviour on the GPU. Based on what the profiling shows, we will explore a few common optimization directions, which may include:

- Combining multiple small GPU operations into fewer, larger ones to reduce overhead.
- Reducing the precision of the model's computations (quantization) to save memory and time.
- Improving how the model uses GPU memory during inference.
- Looking into the attention mechanism to see if it can be made more efficient.

The exact optimizations we apply will depend on what the profiling actually shows to be the bottleneck, and may evolve as we go deeper into the project.

Fallback plan: If we are unable to get meaningful results while working with Qwen2-VL, we plan to switch to Moondream2, another open-source vision-language model, and repeat the same process of profiling and optimization on it instead.

Expected contribution:

Overall, we expect to come out of this project with a clearer understanding of what actually slows down vision-language model inference, and with at least one working, measured optimization that improves speed and/or energy efficiency, along with the profiling evidence to support it.

3. Hardware & Software Requirements

Hardware: A single NVIDIA GPU for most of the project, and a Jetson board towards the end for final testing.

GPU specifications: We do not need multiple GPUs, since this project focuses on optimizing inference on a single device rather than training or scaling across GPUs (minimum 16GB VRAM). We would be happy to work with any one of the following, in order of preference — please provide whichever is available:

1. NVIDIA A100
2. NVIDIA RTX 4090
3. NVIDIA L4 / L40S
4. NVIDIA RTX A6000

Edge device: Towards the final few weeks of the project, once we have a working optimization, we would like to test and demonstrate our results on an actual edge device, since this project falls under the Applications & Edge AI track. For this, we request a Jetson board, in order of preference:

1. Jetson AGX Orin
2. Jetson Orin NX
3. Jetson Orin Nano

This is only needed for the last couple of weeks and is not required for the main development work.

4. Project Plan

Note: The timeline below is a rough, high-level plan and is expected to change as we learn more during the project.

Week	Planned Activity
1–2	Getting familiar with the model and setting up the environment and hardware access
3–4	Running the model, setting up basic measurements, and profiling to find bottlenecks
5	Evaluation 1 — presenting our findings so far and our plan going forward
6–7	Working on optimizations based on what we find, and testing them
7	Evaluation 2 — sharing a working version with initial results
8	Continuing optimization work, and testing on the Jetson board if things are on track
9	Final Evaluation — final results, demonstration, and report

Division of responsibilities:

At this stage, we plan to split the work equally among all three team members, with each member contributing across all parts of the project — setup, profiling, optimization, and evaluation — rather than assigning fixed roles upfront. As the project progresses and we get a clearer picture of the work involved, we will divide tasks more specifically among ourselves based on what is needed at each stage.

5. Expected Outcome

Expected technical result: We expect to identify the key bottlenecks in the inference process of the chosen model, and to apply at least one meaningful optimization that improves either the speed or the energy efficiency of running the model, backed by proper measurements. If time and resources allow, we also hope to demonstrate this improvement running on an actual edge device (Jetson).

How success will be measured: We will measure success mainly through the quality of our process rather than a specific speedup number — that is, whether we correctly identified the real bottleneck through profiling, whether our optimization is backed by clear before-and-after measurements (speed, memory, and energy usage), and whether the model’s output quality is not affected negatively after optimization.